MILITARY SPECIFICATION

ANTENNA COUPLER GROUP, AN/SRA-49( )

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design, performance, and testing of the high frequency (HF) antenna coupler group, hereinafter referred to as the antenna coupler group, for use aboard surface ships for general communications purposes. The antenna coupler group is designed for receive operations over the 2 megahertz (MHz) through 30 MHz frequency range with a high degree of frequency-selective isolation.

1.1.1 AN/SRA-49A. The antenna coupler group is designed to connect up to a maximum of 20 receivers to two antennas.

1.1.2 AN/SRA-49B. The antenna coupler group is designed to connect up to a maximum of 20 receivers to three antennas.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-S-901 | Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment And Systems, Requirements For |
| MIL-E-16400 | Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For |
| MIL-W-16878 | Wire, Electrical, Insulated, General Specification For |
| MIL-E-17555 | Electronic And Electrical Equipment, Accessories, And Repair Parts; Packaging And Packing Of |
| MIL-F-19207 | Fuseholders, Extractor Post Type, Blown Fuse Indicating And Nonindicating, General Specification For |
| MIL-E-21981 | Electronic Type Designations, Identification Plates And Markings: Requirements For |

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

| | |
|---|---|
| SHIPS-R-4984 | Receiving Multicoupler Groups |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures And Tables For Inspection by Attributes |
| MIL-STD-108 | Definitions Of And Basic Requirements For Enclosures For Electric And Electronic Equipment |
| MIL-STD-109 | Quality Assurance Terms And Definitions |
| MIL-STD-167-1 | Mechanical Vibrations Of Shipboard Equipment (Type I - Environmental And Type II - Internally Excited) |
| MIL-STD-454 | Standard General Requirements For Electronic Equipment |
| MIL-STD-461 | Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference |
| MIL-STD-462 | Electromagnetic Interference Characteristics, Measurement Of |
| MIL-STD-781 | Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, Naval Electronic Systems Command (ELEX-8111), Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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| | |
|---|---|
| MIL-STD-1310 | Shipboard Bonding, Grounding, And Other Techniques For Electromagnetic Compatibility And Safety |
| DoD-STD-1399, Section 300 | Interface Standard For Shipboard Systems Electric Power Alternating Current (Metric) |

2.1.2 Other Government publication. The following other Government publication forms a part of this specification to the extent specified herein.

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

| | |
|---|---|
| 0967-LP-283-5010 | Handbook, Electromagnetic Shielding Practices, Submarine |

(Copies of specifications, standards, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The antenna coupler group shall be in accordance with MIL-E-16400, to the extent specified herein.

3.1.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.1.2 Operational requirements. Each antenna coupler group shall give reliable performance within the limits specified herein, under conditions of intermittent or continuous operation during exposure to the adverse conditions normally encountered in Naval ship communication service. Mechanical and electrical ruggedness is of paramount importance for assurance of reliable performance without failure.

3.2 Characteristics. Antenna coupler group characteristics shall be as specified in 3.2.1 through 3.2.3.7.

3.2.1 Performance characteristics. Performance characteristics shall be as specified in 3.2.1.1 through 3.2.1.7.2.

3.2.1.1 Input parameters. The antenna coupler group antenna input distribution lines shall be capable of handling 1000 watts (W) root-mean-square (rms) (continuous) of radio frequency (RF) power over the 2 MHz to 30 MHz frequency range. The peak RF voltage capability shall be 2.1 kilovolts (kV) or greater. Each antenna input distribution line shall be terminated in a 50 ohm dummy load capable of continuously dissipating 500 W rms.

3.2.1.2 Operating frequency range. The antenna coupler group shall be capable of operating as specified herein over the inclusive frequency range of 2 MHz to 30 MHz. Each antenna coupler group shall include couplers capable of being tuned over the ranges specified in TABLE I.

TABLE I. Antenna coupler frequency ranges.

| Antenna coupler | Frequency (MHz) low limit | Range nominal | (MHz) high limit |
|---|---|---|---|
| CU-1799/SRA | 1.990 | 2-6 | 6.020 |
| CU-1800/SRA | 3.990 | 4-12 | 12.020 |
| CU-1801/SRA | 9.900 | 10-17 | 17.400 |
| CU-1802/SRA | 16.600 | 17-30 | 30.500 |

3.2.1.3 Passband characteristics. Passband characteristics shall be as specified in 3.2.1.3.1 through 3.2.1.3.3.



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3.2.1.3.1 Insertion loss. Insertion loss is the power ratio at the receiver before and after insertion of the antenna coupler group in the transmission line. Insertion loss shall be within ±2 decibels (dB) of that shown in FIGURE 1.

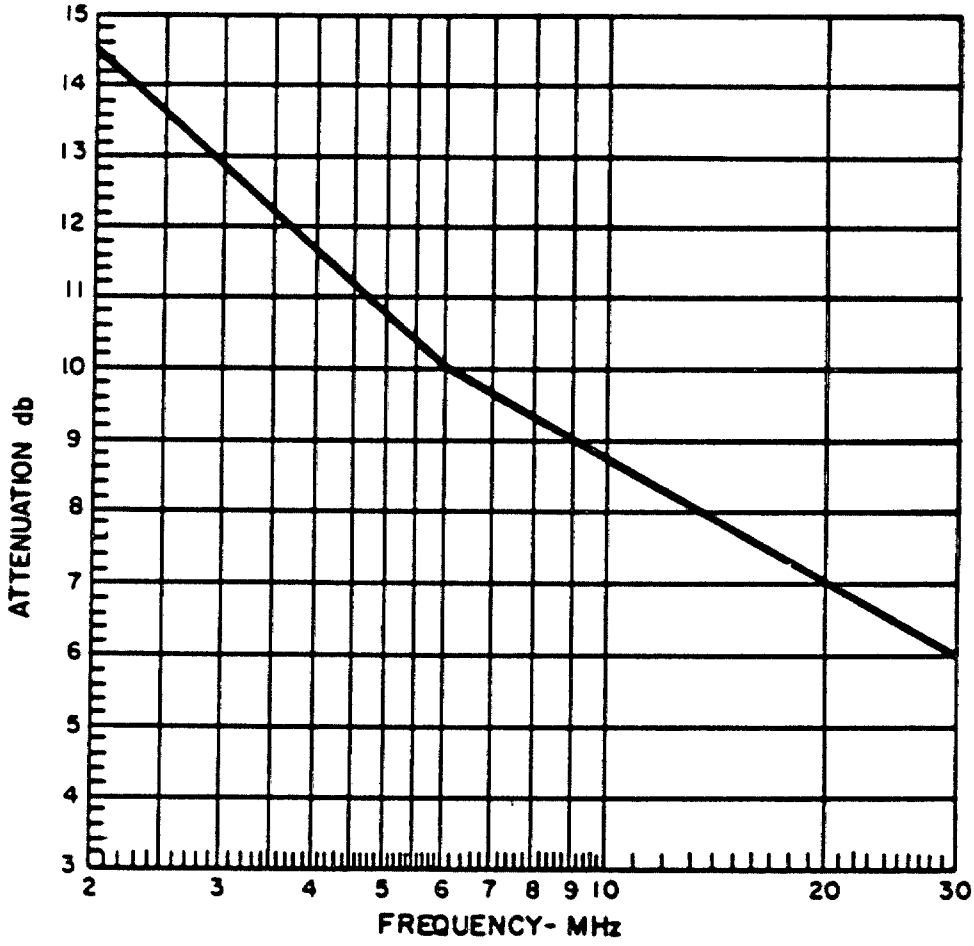



FIGURE 1. Insertion loss.

3.2.1.3.2 Bandwidth. The bandwidth of each coupler unit shall be equal to or greater than 12 kilohertz (kHz) measured at the 3 dB points.

3.2.1.3.3 Ripple. Ripple is variation of signal output within the passband, referenced to the center frequency signal. The ripple within the 3 dB passband shall be 3 dB or less for each coupler unit.

3.2.1.4 Isolation. Isolation is the sum of frequency-selective attenuation and center frequency insertion loss.

3.2.1.4.1 In-band isolation. The isolation of frequencies separated 5 percent or more from passband center shall be equal to or greater than that specified in TABLE II.

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TABLE II. In-band isolation.

| Coupler type | Frequency (MHz) | Insertion loss (dB) | Isolation (dB) |
|---|---|---|---|
| CU-1799/SRA | 2.00 | 14.5 | 80.0 |
| | 3.50 | 12.3 | 74.5 |
| | 6.00 | 10.0 | 69.0 |
| CU-1800/SRA | 4.00 | 11.7 | 73.0 |
| | 7.00 | 9.7 | 68.0 |
| | 12.00 | 8.3 | 63.0 |
| CU-1801/SRA | 10.00 | 8.8 | 64.5 |
| | 13.00 | 8.1 | 62.0 |
| | 17.00 | 7.4 | 59.0 |
| CU-1802/SRA | 17.00 | 7.4 | 59.0 |
| | 23.00 | 6.7 | 56.5 |
| | 30.00 | 6.0 | 54.0 |

3.2.1.4.2 Out-of-band isolation. The isolation at all frequencies in the 2 MHz to 30 MHz frequency range, separated 10 percent or more from center of tuned passband, shall be equal to or greater than that shown on FIGURE 2.

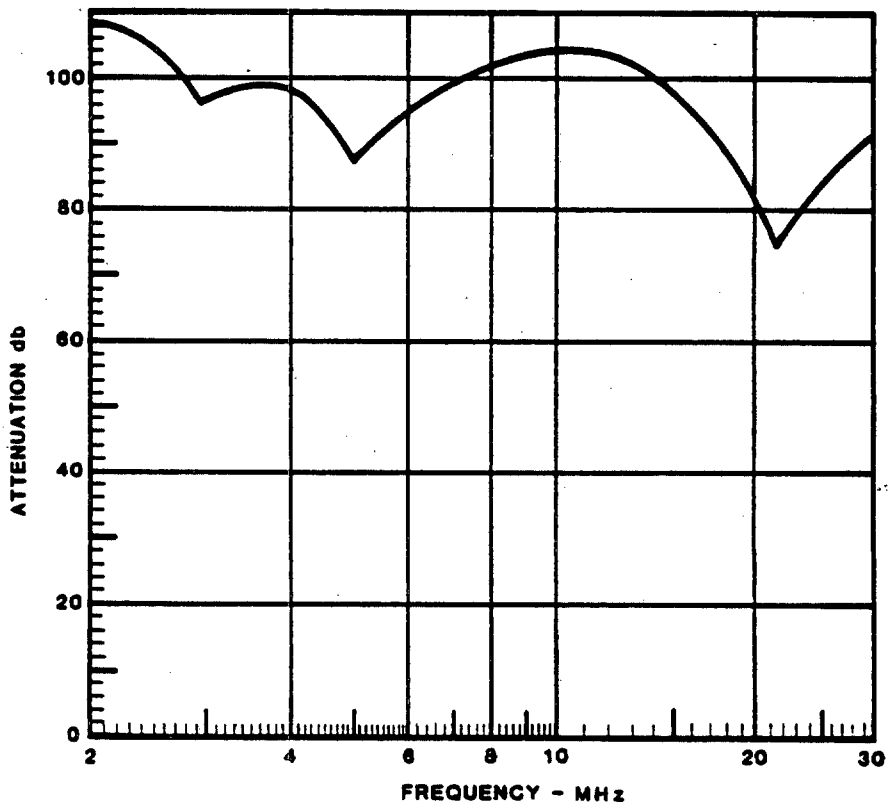



FIGURE 2. Required stop band attenuation vs. frequency.

3.2.1.5 Output load impedance. Each coupler unit shall provide an unbalanced 50 ohm load.


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3.2.1.6 RF overload protection. Each coupler unit shall include overload protection and indication to:

a. Limit coupler unit output voltage to 1.0 volts root-mean-square (V rms) when terminated in 50 ohms

b. Provide visual local and remote indications of overload condition. The overload protection and indication circuit shall be actuated automatically by received RF signals within ±3 percent of the tuned frequency. Overload recovery shall be automatic in not less than 800 milliseconds and not more than 1 second after removal of overload condition.

3.2.1.7 Noise generation. Each coupler unit will be equipped with a random noise generator to facilitate tuning. It shall not be possible to connect the noise generator to the coupler unit unless the coupler unit has been decoupled from the antenna.

3.2.1.7.1 Noise level. The noise level from a coupler unit with noise generator operating shall be 20 dB or more above the ambient noise level of the coupler unit.

3.2.1.7.2 Noise interference. The operation of a coupler unit noise generator shall not generate noise interference levels greater than 3 dB more than ambient noise level in any other coupler unit of the antenna coupler group.

3.2.2 Physical characteristics. Physical characteristics of the antenna coupler group shall be as specified in 3.2.2.1 through 3.2.3.7.

3.2.2.1 Weight. The weight of the antenna coupler group shall be as specified in 3.2.2.1.1 and 3.2.2.1.2.

3.2.2.1.1 Weight (excluding dummy load). The antenna coupler group (excluding dummy loads) shall have a nominal maximum weight of 500 pounds (lbs), exclusive of packing, crating, and maintenance and installation kits.

3.2.2.1.2 Dummy load. Each dummy load shall have a maximum nominal weight of 20 lbs, exclusive of packing and crating.

3.2.2.2 Dimensions. The size of the antenna coupler group shall be specified in 3.2.2.2.1 and 3.2.2.2.2.

3.2.2.2.1 Antenna coupler group (excluding dummy loads). The antenna coupler group (excluding dummy loads) shall have maximum dimensions of 72 inches (in.) high, 15 in. deep, and 29 in. wide, providing for interchangeability of CU-1799, CU-1800, CU-1801, and CU-1802.

3.2.2.2.2 Dummy loads. Dummy loads shall have maximum dimensions of 8.5 in. high, 17 in. deep, and 5.3125 in. wide.

3.2.3 Environmental service conditions. Environmental service conditions of the antenna coupler group shall be as specified in 3.2.3.1 through 3.2.3.7.

3.2.3.1 Operating temperature range. The antenna coupler group shall have the operating temperature range specified in MIL-E-16400 for equipment used in a sheltered controlled environment (Range 4).

3.2.3.2 Nonoperating temperature range. The antenna coupler group shall have the nonoperating temperature range specified in MIL-E-16400 for equipment used in a sheltered controlled environment (Range 4).

3.2.3.3 Humidity. The antenna coupler group shall maintain the performance specified herein when exposed to the relative humidity ranges of MIL-E-16400.

3.2.3.4 Shock. The antenna coupler group shall be capable of withstanding Grade A, Class I, Medium weight, Type A shock test for shipboard equipment of MIL-S-901.

3.2.3.5 Vibration. The antenna coupler group shall withstand the Type I vibration test of MIL-STD-167-1 at an upper vibration frequency limit of 50 hertz (Hz). Vibration test frequencies shall be selected randomly, not in incremental steps.



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3.2.3.6 Inclination. The antenna coupler group shall be capable of withstanding the inclination test of MIL-E-16400.

3.2.3.7 Salt fog. The antenna coupler group shall withstand the salt fog test for sheltered equipment of MIL-E-16400.

3.3 Parts, materials, and processes. Parts, materials, and processes shall be as specified in 3.3.1 through 3.3.3.3.

3.3.1 Selection. Parts, materials, and processes used to manufacture the antenna coupler group shall conform the the requirements of MIL-E-16400, except as specified herein.

3.3.1.1 Nonstandard parts. Requests for approval for the use of nonstandard parts shall be submitted in accordance with MIL-E-16400.

3.3.1.1.1 Nonstandard part documentation. All nonstandard parts shall have been documented in accordance with MIL-E-16400, and approved, prior to conducting first article tests (see 4.3).

3.3.1.2 Previous nonstandard part approval. Government-furnished equipment (GFE), if provided, may include nonstandard parts. The equipment furnished in accordance with this specification shall employ standard parts or approved nonstandard parts. Nonstandard part approvals previously granted under earlier contracts for equipment represented by the GFE shall not be construed as being extended to equipment furnished in accordance with this specification. Nonstandard parts cited in Government furnished drawings shall not be construed as being approved for use in equipment furnished in accordance with this specification.

3.3.1.3 Use of ferrous material and nonlinear parts. No ferrous material or nonlinear part shall be used in the main RF signal path. Nonlinear parts shall not be connected to the input resonator for use with overload detection/protection.

3.3.2 Parts. Parts shall be as specified in 3.3.2.1 through 3.3.2.3.

3.3.2.1 Tuning capacitors. All tuning capacitors shall have an air dielectric and shall conform to Requirement 2 of MIL-STD-454. Tuning capacitors shall have midline or straightline frequency tuning characteristics. Alignment provisions shall be incorporated for each tuning capacitor.

3.3.2.2 Electrical connectors. Electrical connectors shall be provided in accordance with and shall conform to the requirements of MIL-E-16400. Mating connectors shall be provided with cable clamps.

3.3.2.2.1 RF input and output connectors. The coaxial RF connectors for the receiving antenna inputs and receiver antenna output shall be Type C. The receptacles shall be mounted at the top of the equipment cabinet and shall be suitably labeled. The center pin of the Type C connectors shall be made captive in all directions. RF inserts, if used, shall be constrained from nonperpendicular movement to ensure proper mating. All connectors shall be of the solder-lug type; crimped pin types shall not be used.

3.3.2.2.2 Multiconductor connectors. External multiconductor cabling connectors shall be provided in accordance with MIL-E-16400. Mating plugs for multiconductor cable connectors shall be provided with cable clamps. All external mounted connector receptacles shall be provided with a suitable protective plastic cap to prevent damage to the connector when the mating connector is not installed. Connectors shall conform to MIL-E-16400. Separate internal and external connectors shall be used for the application specified in a through c:

a. Primary alternating current (AC) line voltage
b. Receiver audio input
c. Remote overload indicator

3.3.2.2.3 External connectors interchangeability. External connectors, except for connectors used for cables which interconnect units specified in 3.3.2.2.2a, b and c shall be interchangeable with GFE connectors. In the event of any conflict or inconsistency between this interchangeability requirement and other requirements of this specification, the other requirements of this specification shall govern.



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3.3.2.3 Printed circuit boards (PCBs). The fabrication of PCBs and the insertion of parts and assemblies thereon shall conform to the requirements of MIL-E-16400, to the extent of being compatible with the GFE.

3.3.3 Processes. Processes shall be as specified in 3.3.3.1 through 3.3.3.3.

3.3.3.1 Coatings, treatments, and painting. Coatings and treatments shall be as specified in MIL-E-16400. The exterior surface of the enclosure and all front panels shall be painted in accordance with MIL-E-16400, unless otherwise specified herein.

3.3.3.2 Soldering. Soldering shall conform to the requirements of MIL-E-16400.

3.3.3.3 Welding. Welding shall conform to the requirements of MIL-E-16400.

3.4 Electrical design and construction. Electrical design and construction of the antenna coupler group shall be as specified in 3.4.1 through 3.4.5.

3.4.1 Electrical power characteristics. The antenna coupler group shall conform to the interface requirements of DoD-STD-1399, Section 300. Nominal supply voltage shall be 115 volts alternating current (VAC), 60 Hz, single-phase, Type I.

3.4.1.1 Steady-state voltage and frequency. The antenna coupler group shall conform to the maximum dissipation requirements of MIL-E-16400 under the conditions specified in a through f:

a. Normal voltage: 11 VAC
b. Lower voltage limit: 103.5 VAC
c. Upper voltage limit: 126.5 VAC
d. Normal frequency: 60 Hz
e. Lower frequency limit: 54 Hz
f. Upper frequency limit: 66 Hz

3.4.1.2 Transient voltage. The antenna coupler group shall conform to the transient voltage test requirements of MIL-E-16400.

3.4.1.3 Transient frequency. The antenna coupler group shall conform to the transient frequency test requirements of MIL-E-16400.

3.4.1.4 Spike voltage. The antenna coupler group shall be capable of withstanding the spike voltage test of MIL-E-16400.

3.4.2 Phase to ground leakage current. The antenna coupler group shall conform to the requirements of MIL-E-16400 for phase to ground leakage current.

3.4.3 Bonding and grounding. The antenna coupler group shall conform to the requirements of MIL-E-16400 for bonding and grounding and shall include provisions for grounding in accordance with MIL-STD-1310.

3.4.4 Internal wiring and cabling. The antenna coupler group shall conform to the requirements of MIL-E-16400 for wiring and cabling. All wiring shall be polytetrafluoroethylene insulated. Hookup wire shall conform to MIL-W-16878, Type E.

3.4.5 Mounting of parts. The antenna coupler group shall comply with the requirements of MIL-E-16400 for mounting of parts.

3.5 Mechanical design and construction. Mechanical design and construction of the equipment furnished in accordance with this specification shall conform to all pertinent requirements of MIL-E-16400 to the extent of being compatible with the GFE.

3.5.1 Construction. The antenna coupler group shall conform to the requirements of MIL-E-16400 for construction.

3.5.2 Enclosure. The antenna coupler group enclosure shall be in accordance with Requirement 55 of MIL-STD-454 and as specified herein.



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3.5.2.1 Degree of enclosure. The degree of enclosure shall be drip proof in accordance with MIL-STD-108.

3.5.2.2 Rigidity. The enclosure shall comply with the rigidity requirements of MIL-E-16400.

3.5.2.3 Rounded corners and edges. All edges and corners of the enclosure shall be rounded. Sharp edges and points of any kind shall be avoided.

3.5.2.4 Moisture pockets. The antenna coupler group enclosure shall conform to Requirement 31 of MIL-STD-454.

3.5.2.5 Handling. The antenna coupler group shall comply with the handling requirement of MIL-E-16400.

3.5.2.6 External connections. A removable connector panel shall be located on the top of the enclosure for interface connections. The connector panel shall be in accordance with FIGURE 3 or FIGURE 4, as applicable. The removal or insertion of any coupler unit while the equipment is energized shall be possible without disruption of service in other coupler units. Provision shall be made to disconnect automatically the antenna input from the coupler distribution system in the event of a power failure.

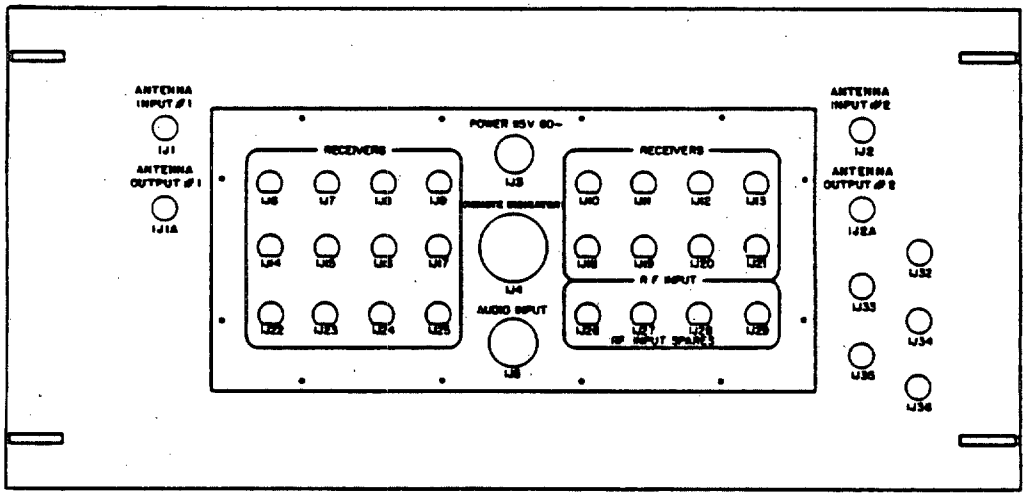
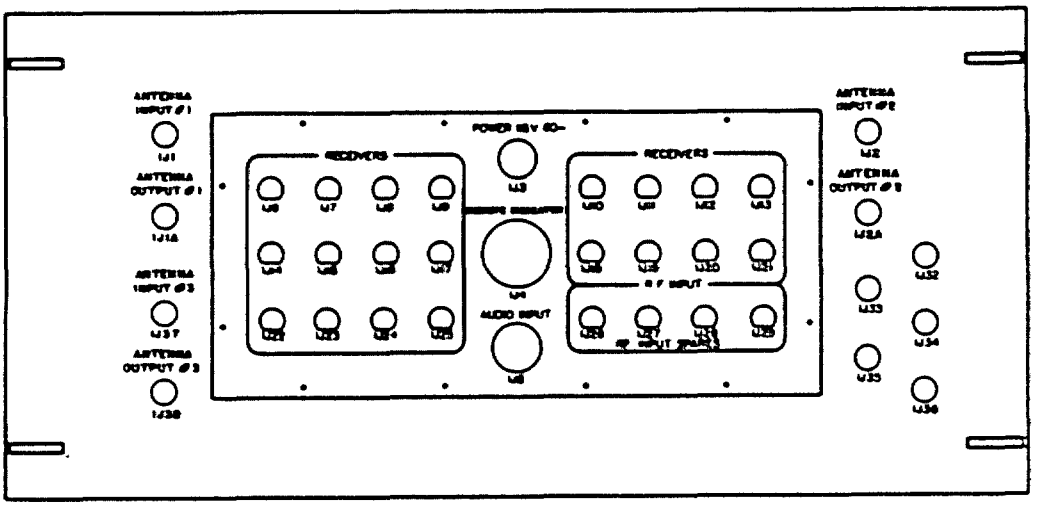



FIGURE 3. Top view of Antenna Coupler Group, AN/SRA-49A.



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| MATING CONNECTORS | | |
|---|---|---|
| J1 & J2 | M39012/06-0002 | UG573C/U |
| J3 | MS 3108R-16-10S | |
| J4 | MS 3108R-28-12P | |
| J5 | MS 3108R-18-1S | |
| | | |
| J6 – J29<br>J32 – J36 | M 39012/06-0002 | |
| AN/SRA-49B<br>J37 & J38 | M 39012/06-0002 | |

FIGURE 4. Top view of Antenna Coupler Group, AN/SRA-49B.

3.5.2.7 Equipment mounting. The antenna coupler group shall be deck-mounted in accordance with MIL-E-16400. Resilient mounts shall not be used.

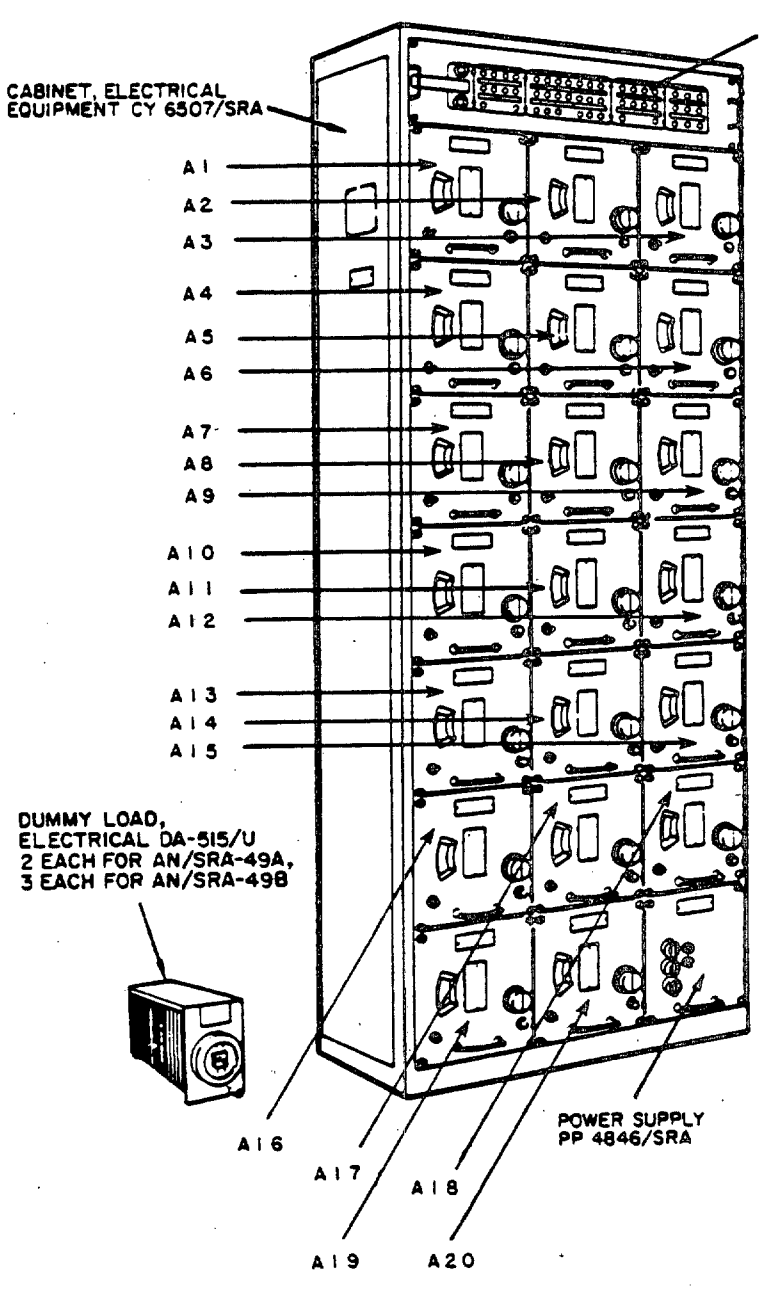
3.5.3 Composition. The antenna coupler group shall consist of 20 individual coupler units, a power supply unit, two AN/SRA-49A or three AN/SRA-49B dummy loads, and a signal distribution panel configured as shown in FIGURE 5. The coupler units and power supply shall be slide-mounted in a cabinet (see FIGURE 6). Withdrawal slides shall be equipped with automatically operated locks which lock the units in both servicing and operating positions. Design differences are specified in TABLE III.

TABLE III. Antenna coupler group design differences.

| Characteristic | AN/SRA-49 | AN/SRA-49A | AN/SRA-49B |
|---|---|---|---|
| Antenna inputs | 1 | 2 | 3 |
| Multicoupler loads | 1 | 2 | 3 |
| Spare RF inputs | 4 | 9 | 9 |
| Audio jack isolation | No | Yes | Yes |
| Low frequency attenuation | No | No | Yes |



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| | AN/SRA-49A | | | | AN/SRA-49B | | | |
|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 |
| A1 | | | | X | | | | X |
| A2 | | | X | | | X | | |
| A3 | X | | | | X | | | |
| A4 | | | | X | | | | X |
| A5 | | X | | | | X | | |
| A6 | X | | | | | X | | |
| A7 | | | | X | | | X | |
| A8 | | X | | | X | | | |
| A9 | X | | | | | | X | |
| A10 | | | | X | | | X | |
| A11 | | X | | | X | | | |
| A12 | X | | | | | | | X |
| A13 | | | X | | | X | | |
| A14 | | X | | | X | | | |
| A15 | X | | | | | | X | |
| A16 | | | X | | | X | | |
| A17 | | X | | | X | | | |
| A18 | X | | | | | | | X |
| A19 | | | X | | | X | | |
| A20 | | X | | | X | | | |

1: CU-1799/SRA
2: CU-1800/SRA
3: CU-1801/SRA
4: CU-1802/SRA

FIGURE 5. Antenna coupler group.

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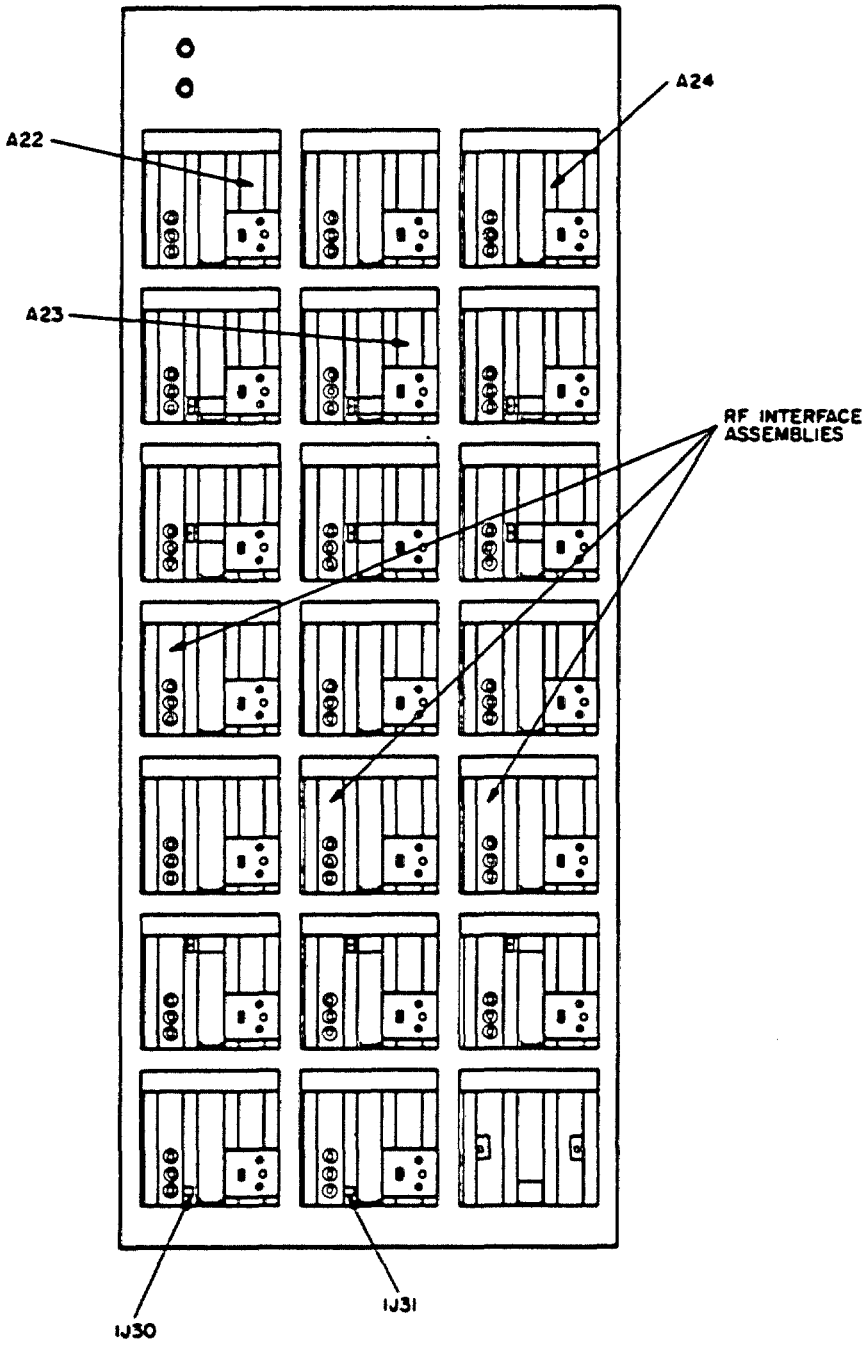



FIGURE 6. Electrical equipment cabinet parts location.

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3.5.3.1 Cabinet. The electrical equipment cabinet shall conform to the requirements of MIL-E-16400. The cabinet is basically the antenna coupler group enclosure, but also provides RF and electrical power interface to the coupler units and the power supply unit.

3.5.3.2 Coupler units. Coupler units shall be interchangeable with the coupler units of the GFE and shall comply with the requirements of MIL-E-16400. The coupler units are basically composed of a front panel (containing tuning control, frequency dial, overload indicator, and noise generator/decoupled switch), a resonator assembly, a PCB assembly, RF interface, power connectors assembly, and a support pin. The design, manufacture, and assembly of gears and cams shall conform to Requirement 48 of MIL-STD-454.

3.5.3.3 Power supply. The power supply unit shall be interchangeable with the power supply unit of the GFE and conform to the requirements of MIL-E-16400.

3.5.3.4 Signal distribution panel. The signal distribution panel shall comply with the requirements of MIL-E-16400 and shall be interchangeable with the GFE signal distribution panel (see 6.4 and FIGURE 7).

3.5.3.5 Dummy loads. Dummy loads shall be in accordance with SHIPS-R-4984. The antenna shall be terminated by the 50 ohm dummy load in the event of power failure.

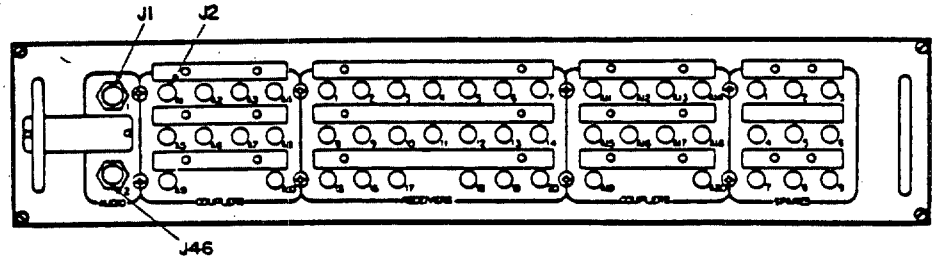



FIGURE 7. Signal distribution panel.

3.5.4 Controls, indicators, and panel layout. The controls, indicators, and panel layout shall be as specified in 3.5.4.1 through 3.5.4.2.5.

3.5.4.1 Controls. Controls shall conform to Requirement 28 of MIL-STD-454.

3.5.4.1.1 Coupler unit controls. Each coupler unit shall be equipped with the controls specified in a and b:

a. Frequency (tuning control)
b. Decoupling/noise generator switch

3.5.4.1.1.1 Frequency control. The frequency control shall be equipped with a locking mechanism. Clockwise rotation (viewed from front panel) of the frequency control shall result in an increase in frequency. The tuning ratio shall be such that the 3 dB bandwidth of each channel can be tuned in not less than 20 degrees and not more than 100 degrees of rotation.

3.5.4.1.1.2 Decoupling/noise generator switch. The decoupling/noise generator switch shall have three positions:

a. Coupled
b. Decoupled
c. Noise generator decoupled (spring loaded to return to decoupled position)

3.5.4.1.2 Power supply controls. The antenna coupler group power supply shall include a power ON/OFF switch.

3.5.4.2 Indicators. Indicators shall be as specified in 3.5.4.2.1 through 3.5.4.2.5.

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3.5.4.2.1 Tuning dial mechanism. Each coupler unit shall be provided with a tuning dial mechanism conforming to Requirement 42 of MIL-STD-454. Tuning dial accuracy and resettability shall be within ±2.0 percent of indicated frequency. Dial windows shall comply with MIL-E-16400.

3.5.4.2.2 Overload indicators. Each coupler unit shall be equipped with overload indicator lights conforming to Requirement 50 of MIL-STD-454.

3.5.4.2.3 Power ON indicator. The antenna coupler group power supply unit shall be equipped with a power ON indicator light conforming to Requirement 50 of MIL-STD-454.

3.5.4.2.4 Fuses. Fuses shall be in accordance with MIL-E-16400, except that extra fuses are not required. Fuses shall be mounted on the front panel, in blown fuse indicating fuseholders, which conform to MIL-F-19207. Spare fuseholders are not required.

3.5.4.2.5 Panel lights. Panel lights shall be easily replaceable from the front panel without requiring removal of the unit.

3.5.5 Threaded and thread-locking devices. Threaded and thread-locking devices shall conform to the requirements of MIL-E-16400, except as otherwise specified herein. All screws, nuts, bolts, and other threaded and thread-locking devices shall be of austenitic corrosion-resistant steel, except in applications requiring material capable of withstanding high tensile stresses. Captive-type bolts shall be used on equipment front panels. Cadmium plating is not acceptable on the captive-type bolts used on the equipment front panels.

3.5.6 Connections and wire terminations. Connections and wire terminations shall conform to the requirements of MIL-E-16400.

3.6 Identification and marking. Identification and marking shall be as specified in 3.6.1 and 3.6.2.

3.6.1 Identification. The antenna coupler group shall be identified as specified in 3.6.1.1 through 3.6.1.4.

3.6.1.1 Item name and type designation. The item name shall be Antenna Coupler Group, AN/SRA-49A or AN/SRA-49B, as applicable, and shall include the units specified in a through g configured in accordance with FIGURE 5.

a. Electrical Equipment Cabinet, CY-6507/SRA
b. Antenna Coupler, CU-1802/SRA
c. Antenna Coupler, CU-1801/SRA
d. Antenna Coupler, CU-1800/SRA
e. Antenna Coupler, CU-1799/SRA
f. Power Supply Assembly, PP-4846/SRA
g. Dummy Load, Electrical, DA-515/U

3.6.1.2 Serial numbers. The antenna coupler group and each unit thereof shall be serialized in accordance with MIL-E-21981.

3.6.1.3 Conductors. Identification of conductors shall conform to the requirements of MIL-E-16400. The method used for identification shall be consistent throughout the equipment.

3.6.1.4 Identification of parts. Identification of all units, assemblies, electrical and moving mechanical parts, and mechanical parts required to be removed for normal maintenance, shall be in accordance wih MIL-E-16400. Hertz or Hz shall be used for all indications of frequency on hardware and associated software. Reference designators shall be the same as used in the GFE to the maximum extent practicable.

3.6.2 Marking. Marking of the antenna coupler group and items thereof shall conform to MIL-E-16400.

3.7 Electromagnetic interference (EMI) and EMC control. The antenna coupler group shall conform to the Part 5 requirements of MIL-STD-461. Publication 0967-LP-283-5010 shall be considered during physical design of internal cable runs, shielding, and connector layout.

3.8 Reliability. The specified lower test mean-time-between failure (MTBF) ($\theta_1$ as defined in MIL-STD-781) for this equipment shall be 1500 hours.

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3.9 Maintainability. The antenna coupler group shall conform to the maintenance design criteria of MIL-E-16400.

3.10 Safety criteria. Safety criteria in accordance with MIL-E-16400, shall be applied during design, selection, and construction of the antenna coupler group.

3.11 Workmanship. Workmanship shall be in accordance with MIL-E-16400.

4. QUALITY ASSURANCE PROVISIONS

4.1. Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through c:

a. First article inspection (see 4.3)
b. Quality conformance inspection
   1. Production inspection (Group A) (see 4.4.1)
   2. Production control inspection (Group B) (see 4.4.2)
   3. Environmental inspection (Group C) (see 4.4.3)
c. Inspection of preparation for delivery (see 4.7)

4.3 First article inspection. Unless otherwise specified (see 6.2), five antenna coupler groups shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE IV.

4.4 Quality conformance inspection.

4.4.1 Production inspection (Group A). Production inspection shall be performed on every antenna coupler group offered for delivery. This inspection shall comprise such examinations and tests which will prove the workmanship and reveal omissions and errors of the production process. Such tests shall, for example, consist of: functional and performance tests at a limited number of points in the required ranges; tests which detect deviations from design; and tests which detect hidden defects of materials. Production inspection shall include the examinations and tests shown in Group A of TABLE IV.

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis in accordance with MIL-STD-105, using inspection level S-3 with an acceptable quality level (AQL) of 6.5 percent for each attribute. Production control inspection shall be conducted on inspection lots that have passed the production inspection of 4.4.1. These tests shall be performed on the complete antenna coupler group as offered for delivery. Production control inspection shall include the examinations and tests shown in Group B of TABLE IV.

4.4.2.1 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defects or screen out the rejected units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from the new lots and shall retain their identity.

4.4.3 Environmental inspection (Group C). Environmental inspection shall consist of the tests specified as Group C in TABLE IV.



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TABLE IV. Examinations and tests.

| Examination or test | Requirement paragraph | Test paragraph | First article | Quality conformance inspection | | |
|---|---|---|---|---|---|---|
| | | | | Group A | Group B | Group C |
| Weight | 3.2.2.1 | 4.5 | X | | X | |
| Parts and materials | 3.3 | 4.5.3 | X | X | | |
| Processes | 3.3.3 | 4.5.4 | X | X | | |
| Dimensions | 3.2.2.2 | 4.5.2 | X | X | | |
| Marking | 3.6.2 | 4.5 | X | X | | |
| Power handling | 3.2.1.1 | 4.5.5 | X | X | X | |
| Frequency range | 3.2.1.2 | 4.5.6 | X | X | X | |
| Passband | 3.2.1.3.1, 3.2.1.3.2, 3.2.1.3.3 | 4.5.7 | X | X | X | |
| Workmanship | 3.11 | 4.6 | X | X | | |
| Isolation | | | | | | |
| In-band | 3.2.1.4.1 | 4.5.8.1 | X | X | | |
| Out-of-band | 3.2.1.4.2 | 4.5.8.2 | X | X | | |
| Overload | 3.2.1.6 | 4.5.9 | X | X | | |
| Noise generator | 3.2.1.7.1, 3.2.1.7.2 | 4.5.10 | X | X | | |
| Electrical | 3.4.1 | 4.5 | X | | X | |
| Steady-state voltage and frequency | 3.4.1.1 | 4.5 | X | | X | |
| Transient voltage | 3.4.1.2 | 4.5 | X | | X | |
| Transient frequency | 3.4.1.3 | 4.5 | X | | X | |
| Spike voltage | 3.4.1.4 | 4.5 | X | | X | |
| Leakage current | 3.4.2 | 4.5 | X | | X | |
| EMI | 3.7 | 4.5.13 | X | | X | |
| Temperature | 3.2.3.1, 3.2.3.2 | 4.5 | X | | | X |
| Humidity | 3.2.3.3 | 4.5 | X | | | X |
| Shock | 3.2.3.4 | 4.5.11 | X | | | X |
| Vibration | 3.2.3.5 | 4.5.12 | X | | | X |
| Inclination | 3.2.3.6 | 4.5 | X | | | X |
| Salt fog | 3.2.3.7 | 4.5 | X | | | X |
| Reliability | 3.8 | 4.5.14 | X | | | X |

4.4.3.1 Sampling for environmental inspection. For the first environmental inspection, one antenna coupler group from the first month's production shall be selected. For subsequent environmental inspections, one antenna coupler group per month shall be selected. Tests shall be conducted on lots that have passed the production inspection of 4.4.1. The production control inspection of 4.4.2 shall be performed after the environmental inspection (see 4.4.3).

4.4.3.2 Noncompliance. If a sample antenna coupler group fails environmental inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results, thereof, and details of the corrective action taken on the process and all units of products which were manufactured with the same conditions, materials, processes, and so forth. If the Government inspector considers that the corrective action will not enable the antenna coupler group to conform to specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. Antenna coupler groups shall not be released for shipment prior to successful resolution of the failure.

4.5 Test methods. The test methods shall be as specified in 4.5.1 through 4.5.14.7.

4.5.1 Examinations and tests. The method of examination or test shall be in accordance with the test methods specified in MIL-E-16400.

4.5.1.1 Test conditions. The basic condition specified in 4.5.1.1.1 shall apply as pertinent to the performance requirements specified herein, except as modified by a particular test or test procedure contained in this specification.

4.5.1.1.1 Ambient temperature. The ambient temperature shall be between 20°C and 30°C.

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4.5.2 Dimensions. The dimensions specified in 3.2.2.2 shall be verified with suitable calibrated measuring equipment.

4.5.3 Parts and materials. The antenna coupler group shall be examined to determine conformance to 3.3.1 and 3.3.2.

4.5.4 Processes. The antenna coupler group shall be examined to determine conformance to 3.3.3.

4.5.5 Power handling capabilities test. The antenna coupler group power handling capability shall be tested in accordance with 3.2.1.1.

4.5.6 Frequency range test. The antenna coupler group frequency range shall be tested to determine conformance to 3.2.1.2.

4.5.7 Passband test. Each coupler unit shall be tested to determine conformance to 3.2.1.3.1, 3.2.1.3.2, and 3.2.1.3.3.

4.5.8 Isolation tests. Isolation tests shall be as specified in 4.5.8.1 and 4.5.8.2.

4.5.8.1 In-band isolation test. Each coupler unit shall be tested to determine conformance to 3.2.1.4.1.

4.5.8.2 Out-of band isolation test. The antenna coupler group shall be tested to determine conformance to 3.2.1.4.2.

4.5.9 RF overload tests. Each coupler unit shall be tested to determine conformance to 3.2.1.6.

4.5.10 Noise generation tests. The antenna coupler group shall be tested to determine conformance to 3.2.1.7.1 and 3.2.1.7.2.

4.5.11 Shock tests. The antenna coupler group (completely assembled) shall be tested in accordance with 3.2.3.4. Upon completion of shock testing, the antenna coupler group shall be examined for visual damage and shall be subjected to the production control inspection (Group B) of 4.4.2 to identify any hidden damage resulting from the shock test.

4.5.12 Vibration tests. The antenna coupler group (fully assembled), with exception of dummy loads, shall be tested in accordance with 3.2.3.5.

4.5.13 EMI and EMC control tests. The antenna coupler group shall be tested using method CE03 of MIL-STD-462 to determine conformance to 3.7.

4.5.14 Reliability test. A sample of at least one antenna coupler group shall be subjected to reliability testing in accordance with Test Plan XXC of MIL-STD-781 to demonstrate the lower test MTBF ($\theta_1$ as defined in MIL-STD-781) of 1500 hours. Environmental and operational duty cycles shall be as specified herein.

4.5.14.1 Temperature cycling. Temperature cycling shall be performed between 0°C and +50°C. Time duration at the specified high or low temperature including transition between the two extremes shall be approximately 4 hours.

4.5.14.2 Vibration cycling. A 4 Hz - 33 Hz - 4 Hz vibration sweep, as specified in Appendix B, the Vibration stress (continuous) paragraph of MIL-STD-781 shall be performed throughout the reliability test. The duration of each complete vibration sweep shall be 10 minutes.

4.5.14.3 Input voltage. Input power shall be nominal during the reliability demonstration test.

4.5.14.4 Operation duty cycle. The equipment shall be nonoperating during the first 30 minute period of each 4 hour test period. The equipment shall be on and operating during the remainder of the test. Maximum rated RF power shall be applied to the input.



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4.5.14.5 Performance measurements. At least daily, insertion loss and isolation measurements shall be made to verify proper equipment operation within the constraints specified in a and b.

a. The insertion loss shall be ± 2 dB of that shown in FIGURE 1
b. Isolation shall be equal to or greater than that specified in TABLE I

4.5.14.6 Failure. The procuring activity shall be notified of any failure during reliability testing. Classification of failures shall be made by the procuring activity in accordance with the Failure categories paragraph of MIL-STD-781.

4.5.14.7 Rejected equipment. If equipment is rejected as a result of reliability testing, the contractor shall immediately notify the procuring activity of the cause(s) for rejection and propose a plan for correction of the deficiencies. When such corrections have been approved by the procuring activity and incorporated in the test unit, the reliability test shall be continued as specified herein.

4.6 Workmanship. The antenna coupler group shall be examined for workmanship in accordance with MIL-E-16400.

4.7 Inspection of preparation for delivery. Inspection shall be performed to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The antenna coupler group covered by this specification is intended for use aboard surface ships for general communication purposes.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 4.3
c. Levels of preservation, packaging, packing, and marking (see 5.1)

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The first article should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

6.4 GFE. The contracting officer should arrange to furnish the GFE listed in 3.5.3.4.

Preparing activity:
NAVY-EC

(Project 5895-N504(EC)

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-A-28880 | ANTENNA COUPLER GROUP, AN/SRA-49( ) |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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